



INSTITUTE OF MANAGEMENT & BUSINESS STUDIES
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DEPARTMENT OF INFORMATION TECHNOLOGY

DIT2145 - SOFTWARE DEVELOPMENT PROJECT
FINAL PROJECT (C# Programming)

Common Instructions

- Refer to the instructions given below thoroughly.
- Below are the instructions for preparing a software related to the final practical exam for the subject "SOFTWARE DEVELOPMENT PROJECT" of the Diploma in Information and Communication Technology.
- Develop your software by using following instructions.
- To get the diploma certificate you have successfully complete the software development project.
- Total Marks: 100

- **Using following scenario develop desktop application.**
- **Use Visual studio programming tool and C# programming language.**

Project Overview.

Sarasavi is a popular fully stocked library with a collection of nearly 500 books. These books are available for loan as well as for reference by “Registered Members”, for free of charge. However,

“Registered Visitors” cannot borrow any book instead they can only refer a book. The functions of the Sarasavi library can be categorized as below:

Loan Process, Return Process, Reservation process, Inquiry Process, Book Registration Process and User Registration process

Details about these functions are given below:

1. Loan Process:

A particular book can have many copies available in the library. A *Copy* is a physical book while a *Title* is the class of all books which are identical, i.e. “*Access 2022 all-in-one desk reference for dummies*, by Alan Simpson, Margaret Levine Young and Alison Barrows, ISBN: 978-0-470-03649-5” is a *Title* while the physical book on the bookshelf is a *Copy*.

The borrower collects the copies of books that is required and hand over them to the library counter. The Librarian will check whether the borrower has an overdue of books to be returned because a borrower can borrow only maximum of 5 books from the library at a time. If the borrower exceeded the maximum count, then they cannot borrow until the overdue books are returned.

Then the Librarian will check the status of each copy which indicates whether the copy is “Reference” only where referenced copies are not “borrowable”. If the book is “Borrowable” then the loan will be confirmed and the expected return date will be informed to the borrower. The Librarian has the authority to accept or cancel the request for a loan and a copy of a book can be borrowed for a period of two weeks.

2. Return Process:

The Librarian accepts the return and checks the status of the copy. If the copy is already reserved, the librarian takes steps to inform the member who has reserved the copy.

3. Reservation Process:

On return, the reservations are checked to see if there is an outstanding reservation for the title a copy of which is being returned. If so, a message is displayed and the librarian puts the book (copy) on one side, a notification is generated for the borrower with the oldest reservation for the title and the oldest reservation for the title is deleted.

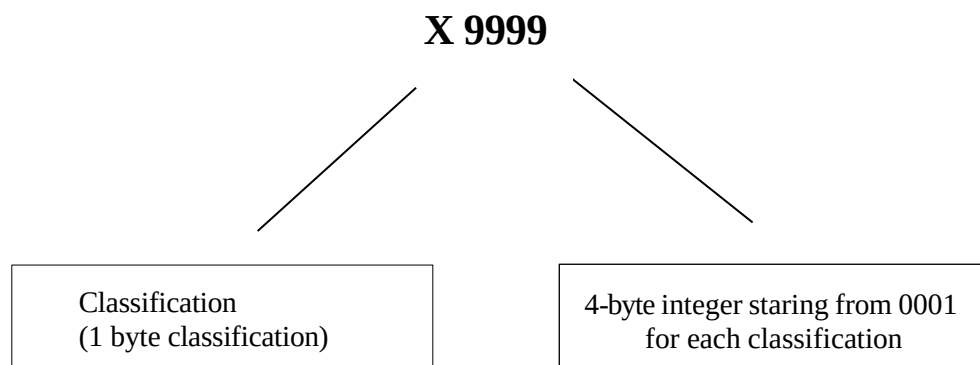
4. Inquiry Process:

The Librarian can also handle inquiries from the borrower about the availability of a book. A facility is also available for a borrower/registered visitor to check the availability of a book. The inquiries may be done by knowing the specific book accession number or knowing a part or whole of the title or author. If it is in the catalogue, Librarian will inform the borrower the status of the book. The status indicates whether the book is available, referenced or not, and in the case of availability of multiple copies, whether all/some/no copies are loaned out, or reserved.

5. Book Registration Process:

The Librarian enters the details of new books and its copies. A maximum of 10 copies are allowed to be registered per book number. The Librarian records the classification, book title, publisher, whether the copy is reference or borrowable.

The book number comprises



The copy number in case of multiple copies has the same structure with an extra number appended at the end.

6. User Registration Process:

New borrowers may also be registered. The following is captured for the user registration. User Number, Name, Sex, National identity card number and Address

Based on the above information given, engineer an object-oriented software system using C#.NET

Assistant Registrar
IMBS GREEN CAMPUS